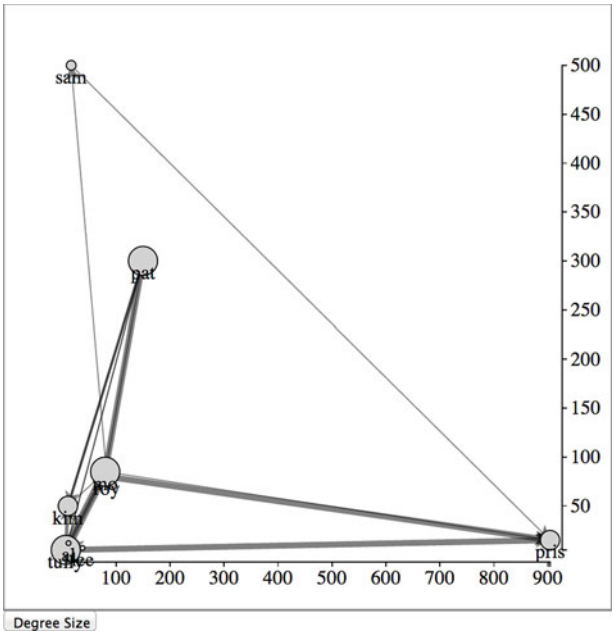


Figure 6.21 When the network is represented as a scatterplot, the links increase the visual clutter. It provides a useful contrast to the force-directed layout, but can be hard to read on its own.

6.2.7 Manually positioning nodes

The force-directed layout doesn't move your elements. Instead, it calculates the position of elements based on the x and y attributes of those elements in relation to each other. During each tick, it updates those x and y attributes. The tick function selects the `<line>` and `<g>` elements and moves them to these updated x and y values.

When you want to move your elements manually, you can do so like you normally would. But first you need to stop the force so that you prevent that tick function from overwriting your elements' positions. Let's lay out our nodes like a scatterplot, looking at the number of followers by the number that each node is following. We'll also add axes to make it readable. You can see the code in the following listing and the results in figure 6.21.

Listing 6.12 Moving our nodes manually

```
function manuallyPositionNodes() {
    var xExtent = d3.extent(force.nodes(), function(d) {
            return parseInt(d.followers)
    });
    var yExtent = d3.extent(force.nodes(), function(d) {
            return parseInt(d.following)
    });
    var xScale = d3.scale.linear().domain(xExtent).range([50,450]);
    var yScale = d3.scale.linear().domain(yExtent).range([450,50]);

    force.stop();
    d3.selectAll("g.node")
         .transition()
```